Evaluating Classifier Performance

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Metrics/measures of Model Performance:

- For y numerical
  - We've seen that RMSE is commonly used
- For y categorical
  - We've seen that accuracy is easily understood (percent correct)
  - Today: let's study other specific measures when y is binary

When y is categorical:

1. The task is called classification
2. The values of y are called labels (or classes)
3. The model can also be called a classifier

Motivating Example

- Sobar (2016) aims to predict y = cases of cervical cancer using x = non-invasive social and behavioral assessments
- They recorded 18 different assessment scores for $n = 72$ individuals
  - $n = 21$ with cervical cancer
  - $n = 51$ without cancer
- A k-nearest neighbors model (using $k = 8$) evaluated using LOOCV results in 87.5% classification accuracy

Summarizing Classification Performance

The table below displays the cross-classification of

- The actual status y and
- The cross-validated prediction $\widehat{y}$

of all 72 individuals.

	Predicted Cancer	Predicted No Cancer
Has Cancer	13	8
Doesn't Have Cancer	1	50

Despite being $< 30\%$ of the data, individuals with cancer made up 8 of 9 incorrect classifications.

Summarizing Classification Performance

A perfect classifier would classify the 72 observations as follows...

	Predicted Cancer	Predicted No Cancer
Has Cancer	21	0
Doesn't Have Cancer	0	51

Confusion AKA Contingency Matrices

The table in the previous slide is known as a **confusion matrix** AKA **contingency table**. Note that the *positive class* is determined by the data analyst.

		Predicted condition	
	Total population = P + N	Positive (PP)	Negative (PN)
Actual condition	Positive (P)	True positive (TP)	False negative (FN)
	Negative (N)	False positive (FP)	True negative (TN)

Confusion AKA Contingency Matrices

Based on these 4 values: TP = 13, FN = 8, FP = 1, TN = 50, there are dozens of possible metrics on classifier. Let's focus on 4:

1. Classification accuracy
2. True positive rate (TPR) vs false positive rate (FPR)

Other related metrics: 1. Sensitivity vs Specificity 1. Precision vs recall

Metric 1: Classification Accuracy

In the previous table the classification accuracy is:

$$\frac{13+50}{13+8+1+50} = \frac{63}{72} = 87.5$$

However, this treats the 1 FP and 8 FN as equally problematic. Especially with cancer, false negatives are a lot worse.

Metric 1: Classification Accuracy

Also classification accuracy is not useful when the positive vs negative y classes are severely imbalanced. For example, research by the St Louis Federal Reserve Bank found the delinquency rate across all US loans to be $y = 1.2\%$.

An algorithm that always guesses $\widehat{y} = 0$ would have 98.8% accuracy. Not impressive! In such cases you might want to use a different metric of accuracy (which we won't cover today):

- Balanced accuracy
- F-1 score

Metric 2: True Positive Rate vs False Positive Rate

True positive rate (TPR) vs false positive rate (FPR) both address the question "how many of each observed class are classified as positive?"

- **True positive rate** (TPR), or $\frac{\text{TP}}{\text{Total Positives}} = \frac{\text{TP}}{\text{TP} + \text{FN}}$
- **False positive rate** (FPR), or $\frac{\text{FP}}{\text{Total Negatives}} = \frac{\text{FP}}{\text{FP} + \text{TN}}$

A perfect classifier has a TPR of 1 and a FPR of 0, but there's an inherent trade off between the two quantities.

The TPR in our example is $\frac{13}{13+8} = 61.9\%$ versus FPR $\frac{1}{1+50} = 2.0\%$

Receiver Operating Characteristics (ROC) Analysis

Classification algorithms return a predicted probability $\widehat{p}$ of the positive class $y = 1$

- These $\widehat{p}$ must be converted to a class label $\widehat{y} = 1$, with the typical threshold p^* for binary classification being 0.5
  i.e. predict $\widehat{y} = 1$ when $\widehat{p} \geq p^*$
- This threshold p^* can be changed to manipulate the trade off between TPR and FPR

Receiver Operating Characteristics (ROC) Analysis

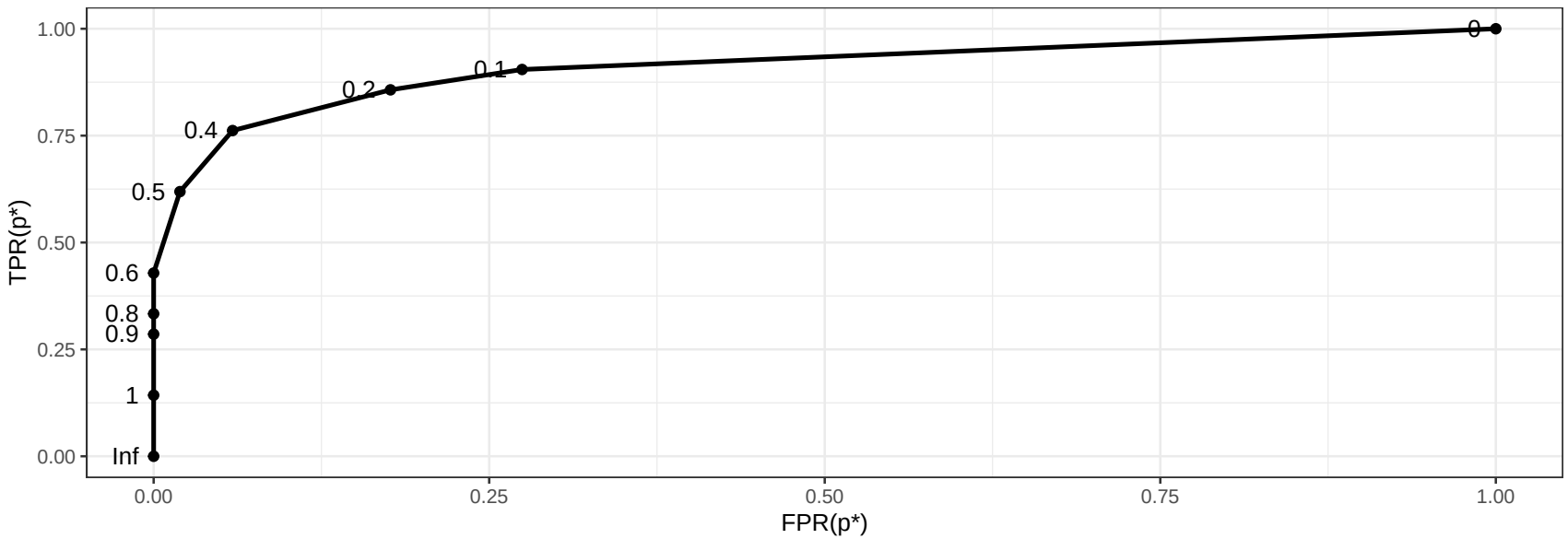
Ex: Say we create a model where we predict $\widehat{y} = 1$ if any observation has at least 3 of it's 8 neighbors have cancer. This results in the following confusion matrix:

	Predicted Cancer	Predicted Healthy
Has Cervical Cancer	16	5
Doesn't Have Cancer	3	48

- The TPR is now $\frac{16}{16+5} = 76.2\%$ versus FPR is $\frac{3}{3+48} = 5.9\%$
- The TPR earlier was $\frac{13}{13+8} = 61.9\%$ versus FPR $\frac{1}{1+50} = 2.0\%$

ROC Curves

The trade-off between $TPR(p^*)$ and $FPR(p^*)$ of a classifier can be summarized by an ROC curve where the dots 0, 0.1, 0.2, . . . , 1 represent all possible values of p^*:



The *area under the ROC curve* (AUC) provides a single-number summary of how well the classifier performs where an AUC value of:

- 0.5 reflects no predictive value
- 1.0 indicates perfect classification
- 0.0 indicates perfect misclassification

SKIP

Skip/ignore remaining slides

SKIP - Other Metrics: Balanced Accuracy

- **Balanced accuracy** addresses class imbalance by averaging the accuracy within each class (ie: averaging the accuracy achieved in each row of the confusion matrix).
- For binary classification, this amounts to:

$$\text{Balanced Accuracy} = \frac{\text{TPR}+(1-\text{FPR})}{2}$$

- Note that this is the average of the TPR and the *true negative rate*.

$$\text{Precision} = \frac{\text{True Positives}}{\text{Total Predicted Positives}}$$

$$\text{Recall} = \text{TPR} = \frac{\text{True Positives}}{\text{Total Positives}}$$

SKIP - Other Metrics: F1 Score

The **F1 Score** is a popular metric that combines a classifier's precision and recall by taking their harmonic mean:

$$F1 = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$

Since F1 focuses on the prediction of the positive class, it tends to be popular in applications that seek to identify relatively uncommon events.

- Examples include: predicting loan default, classifying spam emails, etc.

SKIP - Imbalanced classes, which metric to use?

Consider the following confusion matrix:

Pos	Neg
10	10
5	995

- Classification accuracy is 98.5%
- Balanced accuracy is 74.75%
- The F1-score is 0.571

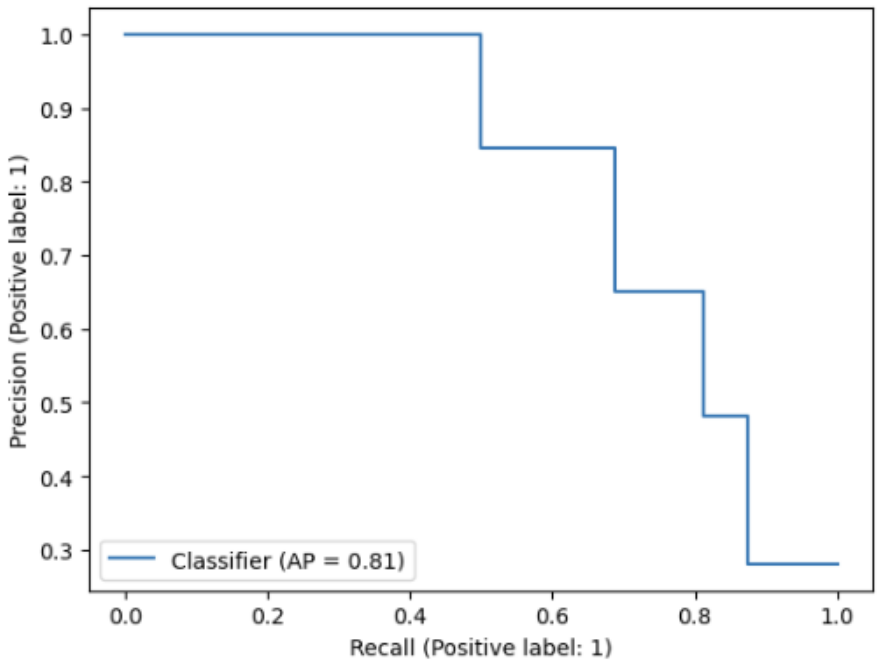
Which metric provides the most useful assessment of this classifier?

SKIP - Imbalanced classes, which metric to use?

- Classification accuracy is acceptable when classes are roughly balanced, and false positives/negatives are equally problematic. It is also the most easily interpreted metric, so non-technical clients may prefer it
- Balanced accuracy is useful when classes are imbalanced, and false positives/negatives are equally problematic
- The F1-score and PR analysis are useful when classes are imbalanced, and you mostly care about predicting the positive class
- ROC analysis and AUC are useful when classes are imbalanced, and false positives/negatives have differential impact

SKIP - Other Metrics: PR-AUC

Precision and recall can also be explored at a variety of classification thresholds in a PR curve:



The area under this curve, known by *PR-AUC* or *AP*, is an alternative single number summary of a classifier's performance.

SKIP - Multiple Classes

What can you do when y is categorical with more than 2 possible levels aka classes aka labels? Extensions to multi-label classification requires adopting *one-vs-rest scheme* (combining many classes to form the "negative" class) or a *one-vs-one scheme* (pairwise comparisons)

	Pr Setosa	Pr Versicolor	Pr Virginica
Setosa	50	0	0
Versicolor	0	48	4
Virginica	0	2	46

- Under a one-vs-many scheme:
  - For Versicolor flowers, the TPR is 48/52 and the FPR is 2/98
- Under a one-vs-one scheme:
  - For Versicolor flowers compared to Virginica flowers, the TPR is 48/52 but the FPR is 2/48

SKIP - Multiple Classes

For multi-label applications there are two popular for calculating single number metrics like AUC or the F1-score:

1. *Micro-averaging* - aggregate the contributions from each class when calculating the metric (only applicable in the one-vs-many scheme)
2. *Macro-averaging* - calculate the metric independently for each class, then take the average (applicable for both one-vs-many and one-vs-one schemes)

For the Iris example:

1. The *micro-averaged TPR* is $\frac{\sum_{j=1}^{k} \text{True Pos.}}{\sum_{j=1}^{k} \text{Total Pos.}} = \frac{50+48+46}{50+52+48} = 0.9600$
2. The *macro-averaged TPR* is $\frac{1}{k}\sum_{j=1}^{k} \text{TPR}_j = \frac{(50/50)+(48/52)+(46/48)}{3} = 0.9605$